Reproduction of Table I of arXiv:1708.02243

Knapen, Lou, Papucci, Setford, Phys. Rev. D 96, 115015 (2017) · 16 October 2026 · production samples 2000 events each · EW LO sample: first 2943 of 3000 events (run in progress)

Questions to answer

- **Q1. Why is the converted-photon upper limit the same for both cross sections?** → Section 7.
- **Q2. Figure 2: why does our |L| go to zero at every second meeting while the paper's does not?** → Section 3 (with Appendix A of arXiv:2002.07503).
- **Q3. Which smearing was applied, and how?** → Section 1.
- **Q4. Which uncertainties are quoted?** → Section 1 (formula) and every table (k/n = ε ± σ).

1. How the hits are smeared, and the uncertainty used

Detector part	Formula applied to the true crossing point	Value
Pixel layers (4)	hit moved to the centre of its pixel: $x_{meas} = p \cdot \mathrm{round}(x / p)$	p = 50 μm (rφ); 250 μm (IBL z), 400 μm (other z)
SCT layers (4), rφ	$(r\varphi)_{meas} = r\varphi + \sigma_{r\varphi} \cdot g$, g = Gaussian random number (mean 0, width 1)	$\sigma_{r\varphi}$ = 34 μm (2 × 17 μm)
SCT layers (4), z	$z_{meas} = z + \sigma_z \cdot g$	σ_z = 0 (“no smearing”) or 1.16 mm (2 × 580 μm)
Beam spot	all hits of an event shifted by the same z_0, z_0 = 45 mm · g	45 mm
Neighbouring pixels	hits in adjacent pixels of one layer merged into one hit	—

Pixel rounding gives an RMS error p/√12 (14 μm for 50 μm, 115 μm for 400 μm). Everything is from the paper (Sec. III A) and Ref. [27], except σ_z = 0 (our choice: it is the only value that reproduces the paper's numbers) and one common z_0 per event (our reading).

Uncertainty of an efficiency ε = k / n: $\sigma = \sqrt{\varepsilon\,(1-\varepsilon)\,/\,n}$ (statistical only)

Decision: **consistent** with the paper if $|\varepsilon - \varepsilon_{paper}| \le 2\,\sigma_0$, $\sigma_0 = \sqrt{\varepsilon_{paper}\,(1-\varepsilon_{paper})\,/\,n}$

2. Table I

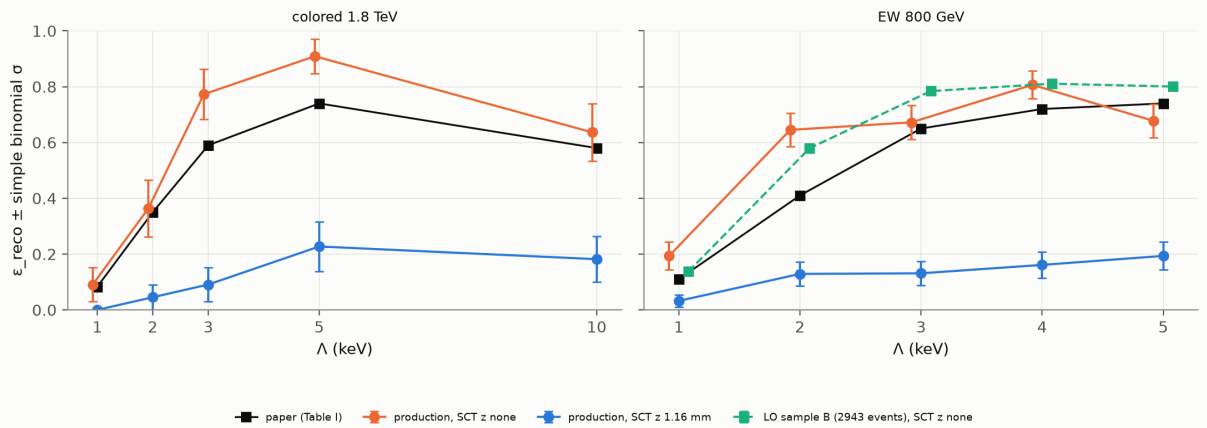
2.1 Efficiencies at every Λ

Colored quirk, m = 1800 GeV (2000 events generated, 873 after matching)

Λ (keV)	ε_{trig} = N(p_T(QQ) > 200 GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig., both charged)	ε_{reco} = N(signal region) / N(fiducial), no smearing	paper ε_{reco}	decision	ε_{reco}, SCT z 1.16 mm	decision
1	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	2/22 = 0.09 ± 0.06	0.083	consistent	0/22 = 0.00 ± 0.00	consistent
2	same	22/68 = 0.32 ± 0.06	8/22 = 0.36 ± 0.10	0.35	consistent	1/22 = 0.05 ± 0.04	**inconsistent**
3	same	22/68 = 0.32 ± 0.06	17/22 = 0.77 ± 0.09	0.59	consistent	2/22 = 0.09 ± 0.06	**inconsistent**
5	same	22/68 = 0.32 ± 0.06	20/22 = 0.91 ± 0.06	0.74	consistent	5/22 = 0.23 ± 0.09	**inconsistent**
10	same	22/68 = 0.32 ± 0.06	14/22 = 0.64 ± 0.10	0.58	consistent	4/22 = 0.18 ± 0.08	**inconsistent**
paper	0.24 → consistent	0.28 → consistent					

EW quirk, m = 800 GeV (2000 events generated, 1045 after matching)

Λ (keV)	ε_{trig} = N(p_T(QQ) > 200 GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig.)	ε_{reco} = N(signal region) / N(fiducial), no smearing	paper ε_{reco}	decision	ε_{reco}, SCT z 1.16 mm	decision
1	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	12/62 = 0.19 ± 0.05	0.11	**inconsistent**	2/62 = 0.03 ± 0.02	consistent
2	same	62/186 = 0.33 ± 0.03	40/62 = 0.65 ± 0.06	0.41	**inconsistent**	8/62 = 0.13 ± 0.04	**inconsistent**
3	same	61/186 = 0.33 ± 0.03	41/61 = 0.67 ± 0.06	0.65	consistent	8/61 = 0.13 ± 0.04	**inconsistent**
4	same	62/186 = 0.33 ± 0.03	50/62 = 0.81 ± 0.05	0.72	consistent	10/62 = 0.16 ± 0.05	**inconsistent**
5	same	62/186 = 0.33 ± 0.03	42/62 = 0.68 ± 0.06	0.74	consistent	12/62 = 0.19 ± 0.05	**inconsistent**
paper	0.10 → inconsistent	0.28 → consistent					



What we learn. Colored: no smearing consistent at every Λ; 1.16 mm inconsistent at Λ = 2, 3, 5, 10 keV. EW: no smearing inconsistent at Λ = 1, 2 keV (higher than the paper), consistent elsewhere; 1.16 mm inconsistent at Λ = 2, 3, 4, 5 keV. The SCT z smearing decides the result: without it our ε_{reco} is close to the paper, with the paper's 1.16 mm it is far below.

2.2 Event counts at every Λ (cut-flow, first crossing)

Colored 1800 GeV: matched 873 → triggered 207 → both charged 68 (same for every Λ).

Λ (keV) →	**1**		**2**		**3**		**5**		**10**	
step ↓ / SCT z →	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**
fiducial (both quirks cross all 8 layers, t < 25 ns)	22	22	22	22	22	22	22	22	22	22
plane found (Δw < 1 cm)	4	1	18	9	21	15	22	19	22	22
+ hit multiplicity (2 hits in 7 layers, ≥ 1 in 8th)	2	0	8	4	17	5	20	9	14	8
+ Δs < 0.01 cm = signal region (pass)	**2**	**0**	**8**	**1**	**17**	**2**	**20**	**5**	**14**	**4**
fail (fiducial − pass)	20	22	14	21	5	20	2	17	8	18
ε_{reco} = pass / fiducial	0.09	0.00	0.36	0.05	0.77	0.09	0.91	0.23	0.64	0.18
$\varepsilon = \varepsilon_{trig} \cdot \varepsilon_{fid} \cdot \varepsilon_{reco}$ (Eq. 12), ×10⁻²	**0.70**	**0.00**	**2.79**	**0.35**	**5.93**	**0.70**	**6.97**	**1.74**	**4.88**	**1.39**
paper ε ×10⁻² (= paper $\varepsilon_{trig}\cdot\varepsilon_{fid}\cdot\varepsilon_{reco}$)	0.56		2.35		3.96		4.97		3.90	

EW 800 GeV: matched 1045 → triggered 186 (same for every Λ).

Λ (keV) →	**1**		**2**		**3**		**4**		**5**	
step ↓ / SCT z →	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**
fiducial (both quirks cross all 8 layers, t < 25 ns)	62	62	62	62	61	61	62	62	62	62
plane found (Δw < 1 cm)	21	14	54	36	61	40	62	45	62	49
+ hit multiplicity (2 hits in 7 layers, ≥ 1 in 8th)	12	7	40	21	41	21	50	24	42	24
+ Δs < 0.01 cm = signal region (pass)	**12**	**2**	**40**	**8**	**41**	**8**	**50**	**10**	**42**	**12**
fail (fiducial − pass)	50	60	22	54	20	53	12	52	20	50
ε_{reco} = pass / fiducial	0.19	0.03	0.65	0.13	0.67	0.13	0.81	0.16	0.68	0.19
$\varepsilon = \varepsilon_{trig} \cdot \varepsilon_{fid} \cdot \varepsilon_{reco}$ (Eq. 12), ×10⁻²	**1.15**	**0.19**	**3.83**	**0.77**	**3.92**	**0.77**	**4.78**	**0.96**	**4.02**	**1.15**
paper ε ×10⁻² (= paper $\varepsilon_{trig}\cdot\varepsilon_{fid}\cdot\varepsilon_{reco}$)	0.31		1.15		1.82		2.02		2.07	

"none" = no SCT z smearing, "1.16" = SCT z 1.16 mm. ε_{fid} = fiducial / 68 (colored, both charged) or 186 (EW). The paper does not quote ε; its value here is the product of its Table I entries.

2.3 EW trigger with the LO sample

EW sample	**ε_{trig}**	**ε_{fid} (4 keV)**	**ε_{reco} at 1 / 2 / 3 / 4 / 5 keV (no smearing)**
Production: MadGraph + Pythia, matched, showered	0.178 ± 0.012	0.333 ± 0.035	0.19 / 0.65 / 0.67 / 0.81 / 0.68
LO, no shower: QQ + 1 jet with $p_T(j) > 200$ GeV (2943 events)	0.096 ± 0.001 (= σ(QQj)/σ(QQ); MG5 2.3.3: 0.096)	0.300 ± 0.008	0.14 / 0.58 / 0.78 / 0.81 / 0.80
Paper	0.10	0.28	0.11 / 0.41 / 0.65 / 0.72 / 0.74

With the LO sample the EW trigger (0.096) and fiducial (0.30) efficiencies match the paper; the reconstruction efficiency does not change. For colored, the same LO ratio gives 0.14 (paper 0.24), and the matched sample matches (0.24).

3. Figure 2 and Appendix A of arXiv:2002.07503

Equations used (the paper's model, solved numerically):

Eq. 3: $d(m\gamma\mathbf{v})/dt = -\Lambda^2 [\sqrt{(1-v_\perp^2)}\,\hat{s} + (v_\parallel/\sqrt{(1-v_\perp^2)})\,\mathbf{v}_\perp] + q\,\mathbf{v}\times\mathbf{B}$ (string + magnetic force)

$L = (\mathbf{x}_1 - \mathbf{x}_2) \times (\mathbf{p}_1 - \mathbf{p}_2)/2$ in the pair's rest frame; $d = |\mathbf{x}_1 - \mathbf{x}_2|$; r = distance of the pair's centre from the beam

Eq. 5: in the pair's rest frame the magnetic field acts as an electric field $\mathbf{E}^* = \gamma_{cm}\,\mathbf{v}_{cm} \times \mathbf{B}$; torque $\boldsymbol{\tau} = q\,\mathbf{d} \times \mathbf{E}^*$

What Appendix A of arXiv:2002.07503 shows (Li, Li, Pei, Zhang, "The quirk trajectory", same equation, uniform magnetic field):

- Eqs. (II.11)–(II.14): in the pair's rest frame the two quirks move back and forth on one line and meet at every integer g ($g = \Lambda^2 t/(2m\rho)$ is the time in units of half an oscillation).
- Text after Eq. (A.4): "the average torque on the system of two quirks is zero when g is increased from 2n to 2n + 2 such that the total angular momentum of the quirk pair is also invariant at g = 2n".
- Eq. (A.5): $z_1(2n) = -z_2(2n) = 0$, i.e. at every second meeting the quirks meet head-on, as at production.

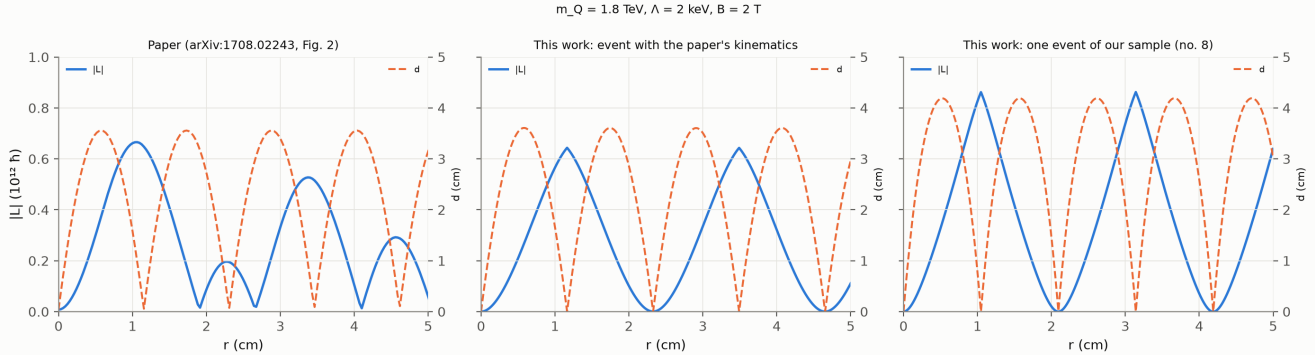
Why. Between two meetings the separation **d** points one way, in the next half-oscillation the other way. The field **E*** is constant, so the torque $q\mathbf{d}\times\mathbf{E}^*$ has the same size and the opposite sign in the two halves: L goes up, then back to zero. Our simulation shows exactly this.

Change of L in each half-oscillation (event with the paper's kinematics, units 10^{12} ħ)	1st	2nd	3rd	4th
total change of L in our simulation	+0.64	-0.64	+0.64	-0.64
from the magnetic force (torque $q\mathbf{d}\times\mathbf{E}^*$)	+0.56	-0.56	+0.56	-0.56
from the string's own term in Eq. 3	+0.09	-0.09	+0.09	-0.09
paper's Fig. 2 (read from its curves)	**+0.65**	**-0.84**	**+0.71**	**-0.80**

In our simulation each half-oscillation adds or removes the same amount (±0.64), so L returns to zero after every two halves. In the paper the halves are unequal (+0.65, −0.84, +0.71, −0.80), so L never returns to zero at a meeting.

Fig. 2 value	Paper (arXiv:1708.02243)	This work: event with the paper's kinematics[1]	This work: one event of our sample[2]
largest separation d (cm)	3.56	3.61	4.19
distance in r between meetings (cm)	1.15	1.16	1.05
|L| at meetings 1–4 (10^{12} ħ)	0.65 / 0.19 / 0.51 / 0.29	0.64 / 0.00 / 0.64 / 0.00	0.86 / 0.00 / 0.86 / 0.00
match with the paper	—	d and first |L|: yes; later |L|: no	pattern as in the middle column

[1] v* = 0.556 and centre speed 0.0975 c, read from the paper's d curve; charges ±2/3. [2] Event no. 8 of our colored sample (MadGraph + Pythia), chosen because its d curve is the closest to the paper's; charges ±1.



Conclusion. Our Fig. 2 is what the paper's own equation gives; the same result is found analytically in arXiv:2002.07503 (Appendix A) and with an independent code. The paper's later |L| values cannot be obtained from its stated equation; the paper does not give the event's charges and momenta or how L was computed.

4. Does L = 0 mean the quirks annihilate?

Paper (Sec. II, text of Fig. 2): "whenever |L| = 0, the separation between the quirks is large", so they do not annihilate. **This work and arXiv:2002.07503 (App. A):** |L| returns to its production value at every second meeting. So the paper's reason does not hold. Two points show why this still does not mean annihilation:

4.1 Classical L ≈ 0 is not a collision

- Classically L = b × p, where b is the distance by which the two quirks miss each other (impact parameter): b = |L| / p.
- To annihilate, the quirks must overlap at the quantum level, $b \sim 1/m_Q = 1.1\times10^{-17}$ cm for m_Q = 1800 GeV. At that distance L is only about one ħ ($b > 1/m_Q$ needs $|L| > 0.67$ ħ).

- In our simulation (and in the independent code) the left-over |L| at the even meetings is 10^6–10^8 ħ, so the quirks still miss each other by millions of times $1/m_Q$:

time steps per oscillation	|L| at meeting 2 (ħ)	|L| at meeting 4 (ħ)	b = |L| / p (cm)	b / (1/m_Q)
3,000	4.2×10^8	1.3×10^8	2×10^{-9} – 7×10^{-9}	2×10^8 – 6×10^8
10,000	5.9×10^6	2.4×10^7	1×10^{-10} – 4×10^{-10}	9×10^6 – 4×10^7
30,000	5.5×10^7	5.8×10^7	9×10^{-10} – 9×10^{-10}	8×10^7 – 9×10^7
100,000	5.5×10^6	2.1×10^6	3×10^{-11} – 9×10^{-11}	3×10^6 – 8×10^6

Event with the paper's kinematics, p* = 1204 GeV. A classical simulation cannot resolve L of a few ħ: the left-over L changes with the time step and does not converge, so we do not know where it would stop.

4.2 Even if L returns exactly to its production value, annihilation is not certain

- arXiv:2002.07503 (App. A, Eq. A.5): at every second meeting the quirks meet head-on, as at production, where the pair's angular momentum is zero (arXiv:2108.06748, Sec. 4: “the initial L'_{tot} at the IP is zero”). With this symmetry annihilation becomes **possible**.
- Possible is not the same as happening. The annihilation cross section of heavy quirks is small, $\sigma v \propto \alpha^2/m_Q{}^2$ (arXiv:2404.13814, Eq. 4.11). Worst case, if an annihilation chance $\approx \alpha^2\gamma v$ is allowed at every second meeting (α = 0.02 / 0.05), the fraction of pairs surviving inside the tracker is 0.995 / 0.970 (colored, 2 keV), 0.971 / 0.835 (EW, 4 keV) and 0.887 / 0.494 (colored, 10 keV): small at the benchmarks, not small at large Λ.
- The symmetry is not perfect in a real detector. arXiv:2108.06748 (Sec. 4) shows that ionisation in material gives the pair angular momentum, and L changes sharply when the pair enters or leaves material, or when only one quirk is inside it. Tracker layers are material. Radiation and a non-uniform field are also absent from the paper's model and ours. (The turning angle η of arXiv:2002.07503, Eq. A.6, rotates the oscillation but keeps the head-on meetings of Eq. A.5, so η alone does not stop L from returning.)
- arXiv:2404.13814 models annihilation only after the pair has radiated its energy and dropped into the quirkonium ground state, not during the oscillations. It finds that radiative energy loss matters only for $\Lambda \gtrsim 100$ keV, far above the keV values here.

Conclusion. L returning to zero (this work, Appendix A) removes the paper's argument, but it does not show that the quirks annihilate. That needs a quantum estimate of the per-meeting probability, which neither the paper nor this study provides. Table I assumes no annihilation.

5. Quirks crossing the same layer more than once

Sample (fiducial events, all crossings within 25 ns)	events with > 2 hits on at least one layer	fraction ± σ
Colored 1800 GeV, 2 keV	13 / 22	0.59 ± 0.10
EW 800 GeV, 4 keV	31 / 62	0.50 ± 0.06

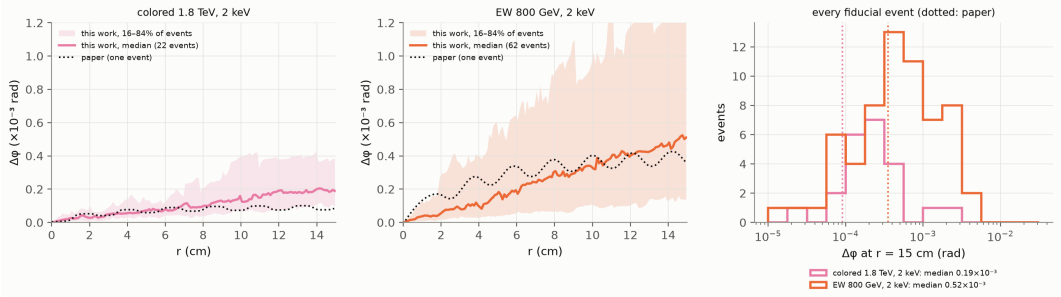
These are results of our simulation. The paper says each quirk “will most likely intersect each layer only once”. arXiv:2002.07503 (Sec. VI, Table I and Appendix D, Table II) also finds repeated crossings, more often at larger Λ and larger m/p_T, with 4–8 hits on some layers at Λ = 2 keV; its percentages (~10%) are for other masses and the CMS tracker and are not directly comparable. Table I above uses the first crossing only.

6. Figures 3, 4, 5

Figure 3: turning of the quirks' plane

Δφ = angle between the plane of the two velocity vectors at radius r and at production (paper's definition)

Fig. 3: Δφ at r = 15 cm (Λ = 2 keV)	fiducial events	ours: median [16–84%]	paper (one event)	our events below the paper's: k / n = fraction ± σ	match
colored 1800 GeV	22	0.19 [0.10, 0.38] ×10^{-3} rad	0.09 ×10^{-3} rad	2 / 22 = 0.09 ± 0.06	**at the low end of our spread**
EW 800 GeV	62	0.52 [0.14, 1.70] ×10^{-3} rad	0.36 ×10^{-3} rad	24 / 62 = 0.39 ± 0.06	inside our 16–84% band



What we learn. The plane turns by less than about 10^{-3} rad over 15 cm in both samples, so the hits stay on one flat plane, as in the paper. The paper's single event lies inside our spread (colored: at its low end).

Figure 4: hits of one signal event

Fig. 4 example event	Δs (plane thickness)	Δw (strip width)	match with the paper
Paper (colored)	83 μm	0.074 cm	—
This work, colored, no smearing	38 μm	0.352 cm	Δs yes (both below the 100 μm cut); Δw no (wider)
This work, EW, no smearing	45 μm	0.100 cm	Δs yes (both below the 100 μm cut); Δw close
This work, colored, 1.16 mm	52 μm	0.470 cm	Δs yes (both below the 100 μm cut); Δw no (wider)
This work, EW, 1.16 mm	73 μm	0.079 cm	Δs yes (both below the 100 μm cut); Δw close

Each example is the signal event with the median Δs. All hits lie on a thin strip, as in the paper.

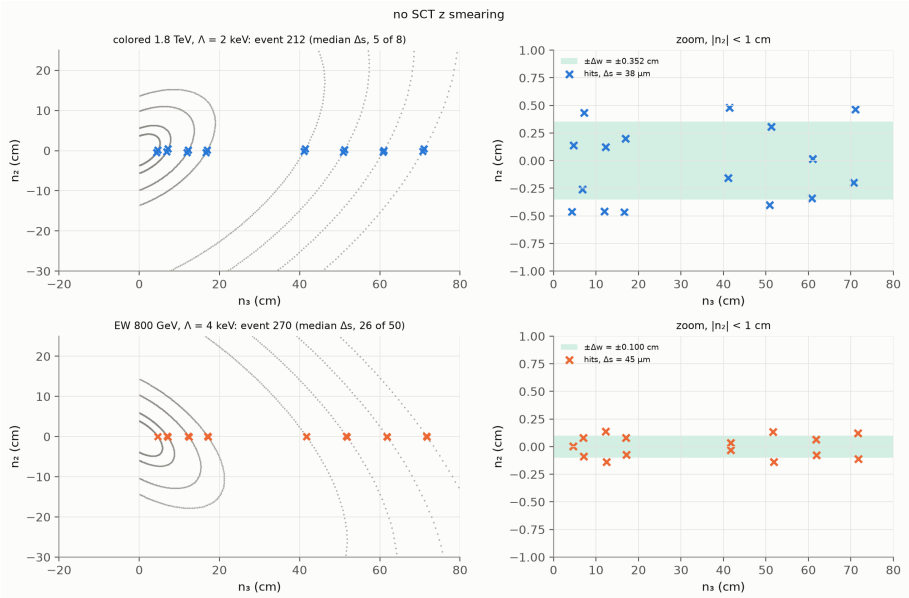
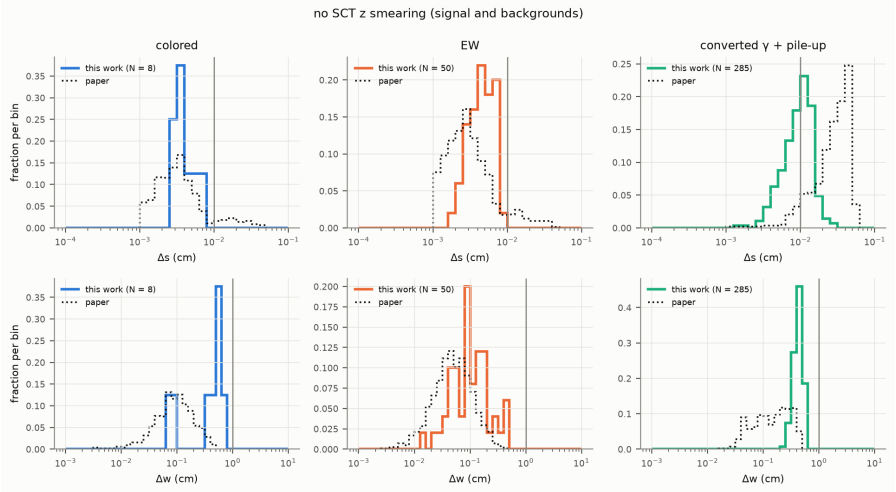


Figure 5: Δs and Δw distributions

Fig. 5	smearing	Δs median: ours / paper	Δw median: ours / paper	fraction with Δs < 0.01 cm: ours / paper	decision
colored signal	none	38 / 32 μm	0.482 / 0.092 cm	8/8 = 1.00 ± 0.00 / 0.90	consistent
colored signal	1.16 mm	160 / 32 μm	0.550 / 0.092 cm	1/4 = 0.25 ± 0.22 / 0.90	**inconsistent**
EW signal	none	44 / 28 μm	0.097 / 0.048 cm	50/50 = 1.00 ± 0.00 / 0.92	**inconsistent**
EW signal	1.16 mm	110 / 28 μm	0.092 / 0.048 cm	10/24 = 0.42 ± 0.10 / 0.92	**inconsistent**
converted γ + pile-up	none	97 / 306 μm	0.438 / 0.156 cm	146/285 = 0.51 ± 0.03 / 0.07	**inconsistent**
converted γ + pile-up	1.16 mm	222 / 306 μm	0.350 / 0.156 cm	0/48 = 0.00 ± 0.00 / 0.07	consistent
pile-up	1.16 mm	317 / 571 μm	0.778 / 0.525 cm	8/109 = 0.07 ± 0.02 / 0.00	**inconsistent**



What we learn. Without smearing the signal Δs (≈ 40 μm) is close to the paper's (≈ 30 μm) and all signal passes the Δs cut (paper ≈ 90%), but 0.51 of the converted-photon planes also pass (paper 0.07). With 1.16 mm no converted photon passes, as in the paper, but most signal fails too. No single SCT z setting reproduces both the paper's signal and its background. Δw is about 5× (colored) and 2× (EW) the paper's; it follows the width expected from our own kinematics (arXiv:2002.07503 Eq. III.1: Δw ≈ belt width/√12).

7. Backgrounds

7.1 How many background events pass and fail

Background sample	SCT z	events	plane found (Δw < 1 cm)	+ hit multiplicity	pass (signal region)	fail	paper
converted γ + pile-up (μ = 50)	1.16 mm	9139	1990	48	**0**	9139	“a few” events at 300 fb^{-1}
converted γ + pile-up (μ = 50)	none	9139	4105	285	**146**	8993	“a few” events at 300 fb^{-1}
pile-up only (μ = 50)	1.16 mm	5000	31 (≥ 4 of 8 layers); 1 (all 8)	—	**0**	5000	none in the signal region

Pile-up planes (5000 events, 1.16 mm)	ours: k / n	rate	paper (approx.)	decision
selected plane, ≥ 1 hit in ≥ 4 of 8 layers	31 / 5000	6.2×10^{-3}	$\sim1\times10^{-3}$	**inconsistent**
selected plane, ≥ 1 hit in all 8 layers	1 / 5000	2.0×10^{-4}	$\sim6\times10^{-5}$	consistent

Pile-up only was not simulated without SCT z smearing. Green: no background in the signal region, as in the paper; red: background in the signal region.

7.2 Converted photons: why the upper limit is the same (Q1)

Converted γ + pile-up (9139 events)	in signal region	per conversion	expected events at 300 fb^{-1}: our σ (0.211 pb) / paper σ (~1 pb)
SCT z 1.16 mm	0 / 9139	$< 3.3\times10^{-4}$ (95% UL)	< 0.21 / < 0.98
no SCT z smearing	146 / 9139	0.0160 ± 0.0013	10 / 48

Answer. The limit is a property of our 9139 simulated conversions: 0 pass, so the 95% upper limit per conversion is $1 - 0.05^{1/9139} = 3.3\times10^{-4}$, whatever the cross section. The cross section only sets how many conversions are expected (300 fb^{-1} × σ × 1% = 633 or 3000), which is why the last column differs. Without SCT z smearing on the background, 146 conversions pass and the background is no longer zero.

8. Open points

- SCT z: the authors' stated resolution gives efficiencies far below Table I; only no smearing reproduces it.
- EW ε_{reco} at Λ = 1–2 keV above the paper (production and LO sample).
- Signal Δw 2–5× the paper's.
- Paper's Fig. 2: unequal half-oscillations; L definition, charges and momenta of its event not given.
- EW LO sample: 2943/3000 events done; showered version pending.

References

Paper: S. Knapen, H. K. Lou, M. Papucci, J. Setford, Phys. Rev. D 96, 115015 (2017), arXiv:1708.02243. [27] ATLAS, ATL-PHYS-PUB-2009-002.
J. Li, T. Li, J. Pei, W. Zhang, “The quirk trajectory”, arXiv:2002.07503 (Eqs. II.11–II.14, A.4–A.5; Sec. VI, App. D).
J. Li, T. Li, J. Pei, W. Zhang, arXiv:1911.02223 (EW trigger 0.1 as an LO ratio, App. A). J. Li et al., arXiv:2108.06748. J. L. Feng et al., arXiv:2404.13814.
Data and scripts: R_data_2026_10_15.py (all numbers), R_report_2026-10-16_v3.py, F38/F39/F40 figures; decision log D17–D44.